WORKSHEET 6.6

Properties of Definite Integrals

Unit 6: Integration and Accumulation of Change

Key ideas

- $\int_a^b f\,dx + \int_b^c f\,dx = \int_a^c f\,dx, \quad \int_b^a f\,dx = -\int_a^b f\,dx, \quad \int_a^a f\,dx = 0.$
- Constants factor out, and sums split: $\int_a^b [kf+g]\,dx = k\int_a^b f\,dx + \int_a^b g\,dx.$
- A constant integrates to the constant times the width: $\int_a^b k\,dx = k(b-a).$
- For an odd function, $\int_{-a}^a f\,dx = 0$. For an even function, $\int_{-a}^a f\,dx = 2\int_0^a f\,dx.$

Worked examples

Example 1 · Joining and reversing

Given $\int_1^4 f(x)\,dx = 5$ and $\int_4^9 f(x)\,dx = -2$, find $\int_1^9 f(x)\,dx$ and $\int_9^4 f(x)\,dx$.

Solution. Add for the first; change the sign for the second.

3 **and** 2**.**

Example 2 · A constant inside

Given $\int_0^3 f(x)\,dx = 7$, find $\int_0^3 [2f(x)-4]\,dx$.

Solution. $2(7) - 4(3)$.

2**.**

Example 3 · Symmetry

Evaluate $\int_{-2}^2 \left(x^3 + 3x^2\right)\,dx$.

Solution. x^3 is odd and contributes 0. $3x^2$ is even: $2\int_0^2 3x^2\,dx = 2(8)$.

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Practice

Questions 1 to 6 use: $\int_0^2 f(x)\,dx = 5$, $\int_2^7 f(x)\,dx = 8$ and $\int_0^7 g(x)\,dx = -4$.

CORE

1. Find $\int_0^7 f(x)\,dx$.

2. Find $\int_7^2 f(x)\,dx$.

3. Find $\int_0^7 [f(x) + g(x)]\,dx$.

4. Find $\int_2^2 f(x)\,dx$.

EXAM LEVEL

5. Find $\int_0^7 [3f(x) - g(x) + 1]\,dx$.

6. Find $\int_0^2 [f(x) + 3x]\,dx$.

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7. An odd function h has $\int_0^5 h(x)\,dx = 6$. Find $\int_{-5}^5 h(x)\,dx$ and $\int_{-5}^0 h(x)\,dx$.

8. Evaluate $\int_{-\pi}^{\pi} (\sin x + 2)\,dx$.

Question 9 · Free-response practice

A continuous function f satisfies $\int_{-2}^3 f(x)\,dx = 10$ and $\int_3^6 f(x)\,dx = -4$.

(a) Find $\int_{-2}^6 f(x)\,dx$.

(b) Find $\int_{-2}^3 [f(x) - 2]\,dx$.

(c) Find $\int_6^{-2} f(x)\,dx$, and find the value of the constant k for which $\int_3^6 [f(x) + k]\,dx = 5$.

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Worked solutions

1. $5 + 8 = 13$.
2. -8.
3. $13 + (-4) = 9$.
4. 0.
5. $3(13) - (-4) + 1(7) = 50$.
6. $5 + \displaystyle\int_0^2 3x\,dx = 5 + 6 = 11$.
7. 0 and -6.
8. $\sin x$ is odd; the constant gives $2(2\pi) = 4\pi$.
9. (a) $10 + (-4) = 6$.
   (b) $10 - 2(5) = 0$.
   (c) -6. For the second: $-4 + 3k = 5$, so $k = 3$.

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